



Data Quality and Integrity

Every day in the United States, patients' health data is collected, shared, and used in different ways throughout the healthcare continuum. The health information (HI) professional plays a vital role in managing the vast amount of clinical, financial, or administrative data that is collected and the information derived from this data. The electronic health record (EHR) is the primary repository of the data that is collected over time, which is used to build a patient's longitudinal health record and provide the basis for care that is appropriate for each specific patient. As guardians of the legal health record, HI professionals should focus on maintaining the integrity and quality of the health data collected and safeguard all information in the health record.

To Do:

1. Study the lesson content below
2. Complete the assigned reading
3. Check your understanding using the activity at the end of the presentation

Lesson Topics

- [Master Patient Index](#)
- [Patient Matching Processes](#)
- [Information Systems](#)
- [Data Integrity and Auditing](#)
- [Check Your Understanding](#)

Master Patient Index

When patients present to a hospital or other place of healthcare delivery for healthcare services, their health record must be located or newly created. The master patient index (MPI), which has evolved over time from a classic rolodex to sophisticated computer software, is the means of locating a patient's unique health data. The integrity of the MPI is crucial to providing appropriate patient care.



Required: Within *Introduction to Information Systems for Health Information Technology*, reference Chapter 7: Master Patient Index (4th Ed. pages: 155-158, 5th Ed. pages: 166-168)

Patient Matching Processes

A key element to maintaining data integrity and patient safety is making sure that data collected and retrieved, especially across multiple care settings, is connected to the right patient. Duplication and/or overlay of patient health records compromises data integrity and puts patient safety at risk. Patient matching processes aim to overcome these errors and accurately match patients with their health records from various sources.

Although there are many reasons why patient data may be duplicated, poor data integrity is the biggest underlying culprit and includes circumstances such as:

- Failing to standardize data
- Missing, inaccurate or outdated data
- Changes in name, address, or phone number
- Patients going by more than one name (e.g., Robert/Bob/Bobby)
- Data entry errors (e.g., misspellings, number transposition)

Processes to rectify and prevent health record duplication include:

▼ Cleaning Up the Master Patient Index
Untangling health record duplications or overlays often requires time consuming manual labor. Matching the submission of an organization's records to a third-party Big Data company against the company's robust database of demographic data is an alternative. Besides identifying who records belong to, this process assigns patients a unique "key" that can be added to the Master Patient Index to be used as another attribute for future identification and prevention of duplicate records.
▼ Improving Ongoing Data Integrity
Data integrity can be improved by implementing data standardization policies, using optical character recognition to avoid transcription errors, and using web-based services that offer data lookups of demographic data.
▼ Authenticating Patient Identities
Individuals can create false identities by stealing someone else's or by combining elements from the identity of several other individuals. Checking with a credit bureau is an option for authenticating the identity presented by an individual seeking medical care.
▼ Verifying Authenticated Identities Belong to the Patients
After the identity presented by an individual who is registering for care has been authenticated, it must then be verified by the registrar as belonging to the registering individual. Verification can take several forms: • Inspection of a government issued photo ID (e.g., driver's license, passport) • Posing knowledge-based questions based on information that is not publicly available • Issuance of smart cards or unique patient identifiers • Use of biometrics (e.g., fingerprint, voice patterns)

Information Systems

A healthcare organization handles vast amounts of diverse kinds of data. Multiple information systems are necessary to store, retrieve and integrate this data in a way that supports all functions of the organization. Health Information professionals are directly or indirectly involved in the design, management, and maintenance of most of these systems. Review these three groups of systems and their relationship with health information.



Group #1: Clinical Information Systems

Clinical information systems (CISs) collect and store information related to patient care. The clinical data stored in the CIS is used to diagnose a patient's condition, make treatment decisions, monitor the patient's current condition, and manage overall care. The data stored in clinical information systems is the basis for information in the EHR.

Since HI professionals are not clinicians, they are usually not direct users of clinical information systems. However, due to their expertise in health record management, they are a valuable source of input when these systems are being designed and implemented. They should be consulted to ensure that accreditation, regulatory, and privacy and security issues are addressed as well as HI concerns such as documentation quality, confidentiality practices, retention schedules and EHR integration and compatibility.

Group #2: Patient Registration Information Systems

The patient registration system is an administrative information system that stores demographic information and performs functionality related to registration, admission, discharge, and transfer of patients within the organization.

Although HI is not responsible for patient registration, the information generated from this system is used by the HI department in the following ways:

- The discharge list generated by this system is used to confirm that the health records of all discharged patients have been identified so they may be analyzed and coded.
- Information about a patient encounter such as the admission date, discharge date, and attending physician is used to support coding and clinical documentation integrity (CDI) functions.
- The HI department receives forms printed from this system (e.g., general consent form, notice of privacy practices acknowledgment, and advance directives) so they can be scanned for storage in the electronic health record.

Group #3: Financial Information Systems

The portion of an organization's financial management system that deals with accounts receivable and the accounts payable is called the revenue cycle. Managing the revenue cycle requires the efforts of employees from many departments who review:

- Claims submitted
- Outstanding claims
- Returned claims
- Denied claims
- Missing accounts
- Bill holds

The work of HI professionals is a critical part of managing this process and directly impacts the condition of the revenue cycle through data generated by the following:

▼ Coding
Assignment of ICD-10-CM, ICD-10-PCS, CPT and HCPCS codes is the basis for calculating the reimbursement requested on medical claims. If the coded data submitted on the claim is incorrect or noncompliant, the reimbursement requested will not be appropriate.
▼ Clinical Documentation Integrity (CDI)
Clinical coding can only be as good as the documentation it is founded on. CDI programs seek to improve clinical specificity which allows coding professionals to assign the most specific diagnosis and procedure codes.



▼ Chargemaster

Chargemasters consist of codes and their associated charges for routine services (e.g., lab, radiology). The chargemaster allows the charges for these routine services to be automatically added to a patient's account without direct oversight from a coding professional. Chargemasters must be updated each year to incorporate code changes and charges associated with each code reviewed for accuracy. Failure to update the chargemaster leads to inaccurate coded data being put on patient accounts and lost revenue.

Data Integrity and Auditing

Healthcare audits are used to identify areas of error or noncompliance so that quality of care and control of healthcare related costs can be improved. However, the results of an audit are only as good as the data the audit is founded on. Unreliable data will lead to unreliable audit results and end in defeating the purpose of the audit. The integrity and quality of the data collected by health information professionals is key to a successful audit process.



Within *Introduction to Information Systems for Health Information Technology*, reference Chapter 13: Audit Program (4th Ed. pages: 321-328, 5th Ed. pages: 331-339).

Check Your Understanding

Follow instructions to review each of the six terms and definitions displayed on the flashcards below.

Patient Registration System

Next Card



Next Steps

This concludes your work in Lesson 2! Assess your understanding of the content presented in this module by taking the Module- level Quiz.

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